Optimizing the Stocks and Sales of Mittal Variety Store

Proposal report for the BDM capstone Project

Submitted by

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Declaration Statement

I am working on a Project titled **"Optimizing the Stocks and Sales of a Mittal Variety Store".** I extend my appreciation to **Mittal Variety Store**, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from primary sources and carefully analyzed to assure its reliability.

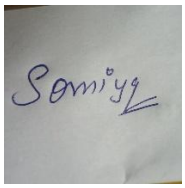
Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfillment in the BS Degree Program offered by IIT Madras. The institution does not endorse any of the claims or comments.

Signature of Candidate:



Name: Somiya

Date: June 2,2025

Executive Summary

The project focuses on the grocery store located at Jaito (Punjab).The Business is a B2C and deals in the essential groceries like biscuits, namkeen, spices and other groceries.

The store is established five years ago by Mr. Rajinder Kumar with the initial investment. Mr. Rajinder Kumar is the sole owner of shop due to which they have to face the challenges to manage the shop using traditional approaches in rising competition. However, the store owner is concerned about inventory management, pricing strategy and credit management. Due to undefined credit limits sometimes owner have to face the loss due to the overdue of payments from customers. To resolve these issues the store will focus on optimizing inventory by analyzing the demand and sales trend and implementing a digital payment method.

The issues will be addressed by analyzing the data via different analytical approaches to obtain a fruitful outcome. Moreover, the project will use tools like Google Sheets, Inventory Management Tracker. The analysis of the data will help us to improve the profit margin, stockouts and excessive stock. The expected outcome includes the resulted inventory system, credit control framework to oversee payments for minimizing missed collections and enhancing cash flow.

Organization Background

Mittal Variety Store is a long-standing, individually retail outlet situated on **Choudhary Chet Singh Street in Jaito, District Faridkot, Punjab**. The store is owned by **Mr. Rajinder Kumar** and has been consistently serving the surrounding community for more than five years. This business operates under the Business-to-Consumer (B2C) category. The store contains a large variety of goods and everyday products, including essential products. The business model follows a traditional approach, which includes the manual inspection of goods and ensuring customer satisfaction before restocking. Billing is done manually, and remaining payments are made in cash, not through a digital platform. All operations are handled by a single person, with very little automation, digital tracking, or without use of modern technology. Due to competition among retailers from organized retail chains and a large number of e-commerce platforms offering variety, the store remains a reliable choice for many local customers, known for its dependable service, convenient access, and diverse product availability.

Problem Statement

3. 1 The store is currently facing challenges in keeping accurate records of its inventory, moving to situations of the stockout and overstock. Efficient Inventory control is especially important for items which have less shelf time and can minimize loss and improves profitability.

3.2 There is a lack of knowledge and experience regarding how to establish prices that are reasonable, profitable, and competitive. It becomes difficult for an owner to attract customers maintaining effective profitability.

3. 3 The absence of an organized credit system is impacting the healthy cash flow of the business. Difficulties such as delayed payments and poor records are common. Establishment of structured credit policies decrease the risk of loss and ensure timely payments.

Background Problem

The Kirana store established in 2020, has been serving to a urban locality for the past five years selling essential groceries, packaged foods and household items. The store has maintain its strong reputation and customer belief by consistently offering best products at reasonable prices. However, the steady flow of customers, store is facing difficulties in inventory management, pricing strategies and credit handling which are impacting profitability and efficiency. Rajinder Kumar, the solo owner, handles the shop using traditional and conventional methods. Inventory is operated based on the estimations and memory-based data, which impacts on the stock management and leads to frequent stockouts of fast-moving items and overstocking of seasonal items or less marketing items. Lack of comprehensive inventory tracking system also causes items to expire, leading to losses. Seasonal changes also complicate the demand of goods making inventory planning inconsistent and less productive. Besides inventory issues the shop is under pressure of reasonable pricing due to increasing competition. Customers demand towards lower prices and discounts increasing regularly forcing the owner to sell at minimal profit or even cost to cost. Without a data driven-models it becomes very difficult to maintain steady profit margins and meet revenue goals. Moreover, Credit sales are

operated manually through handwritten notes in a diary. This outdated system results in missed follow-ups, delayed payments and cash inflow issues. The number of credit transaction increases keeping record becomes more complicated, emphasizing the need for a Structured, Digitalized solution to streamline inventory and credit management for better business performance.

Problem Solving Approach

To solve the challenges faced my Mittal Variety store related to inventory tracking, pricing inefficiencies and unstructured credit system a systematic and structured problem-solving approach should be adopted, initiating with the collection of data and data analysis.

1. Data Collection and Key Questions

- **Stock Records** - Gathering Data of past stock levels, frequency of replenishments, shelf -life data and identification of expired goods.
    - **Key Questions -** What are the most repeatedly sold items and which products often remain unsold? Are there any recurring issues of overstock and shortage of stock
- **Sales and pricing data –** Collection of the daily revenues, purchase costs and margins to assess pricing effectiveness.
    - **Key Questions –** Which Products have better profit margins? How does pricing impact sales? Are you satisfied with the current prices aligned with the market expectations.
- **Credit Transactions -** Collection credit-based sales data and repayment timelines.
    - **Key Questions –** Which customers delay payments and what number revenue tied in the credit? What is the typical repayment cycle.
- **Customer Data-** Gather information on customer demographics, purchasing behaviour, and feedback.
    - **Key Questions –** Who are repeated customers? What do customers prefer to buy in festival seasons and high demand period.

2. Analytical Techniques

a. ABC & Dead Stock Analysis

- Use the ABC & Dead stock analysis to organize the product based on Value and frequency of sale, helping identify which items to focus on and which to eliminate from stock.

b. Demand-Curve Supply Analysis

- Use the **Demand and Supply Curve** to understand the relationship between price and quantity sold. By Assessing the pricing strategy and customer demand we can determine the reasonable pricing that makes balance of affordability with profit margins.

c. Ratio Analysis

- Performs a **Gross Margin Analysis** to determine the profitability of each product line, which helps in identification of products contribute most to the bottom line.
- Calculate the **Receivables Turnover Ratio** indicates prompt collections and health cash flow, while a lower ratio may signal delayed payments or inadequate follow-up on outstanding dues.

d. Data Visualization

- Use of Data Visualization tools like bar charts, graphs and heat map to clearly present the understandings from the analysis. For example, plotting the sales trends over time or visualizing the distribution of stock-outs across different products can highlight the areas for improvement.

3.Problem-Solving Implementation

- **Smart Inventory Control -** Using the ABC & Dead stock analysis to classify products based on sales values and movement. Set a minimum Amount of items to avoid the stockouts and implement expiry tracking for slow moving items.
- **Strategic Pricing Optimization -**Use cost-plus and margin-based pricing to make sure the profitability on each item sold. Regular monitoring of prices according to the competitors to stay market relevant without compromising the profits.
- **Efficient Credit and Collections System**: Implementation of digital payment methods and record customers remaining dues to maintain health cash flows. Set a limit of credit and payments based on customer payment history and purchase behavior.

By strategically applying these analytical tools, Mr. Rajinder Kumar can get a deeper understanding of its concerns, Data-driven decisions and use of modern analytical methods to increase the profitability and reverse its declining sales.

Expected Timeline

6.1 Work Breakdown Structure

- Q &A with Owner (May25-28): Discussion With the store owner to Understood complications.
- Problem Identification (May29-June1): Identification of challenge in inventory, Pricing and credit.
- Data Collection (May 2-7): Collection of data on sales, stock and customer credit.
- Data Cleaning (May08): Organized and cleaned the data for analysis.
- Proposal Preparation (May 9-12): Drafted and finalized project proposal.
- General Insights (June 14-23): Reviewed Data and derived initial observations.
- Mid-term Report (June 25-July06): Prepared the Mid-term project report.
- Solution Building (July 8-19): Analyze Data and work on problem-solving Strategies.
- Final Report (June 21-29): Compiled all findings into the final report.
- Ppt Design (August 03-20): Created Presentation Slides for final Submission.

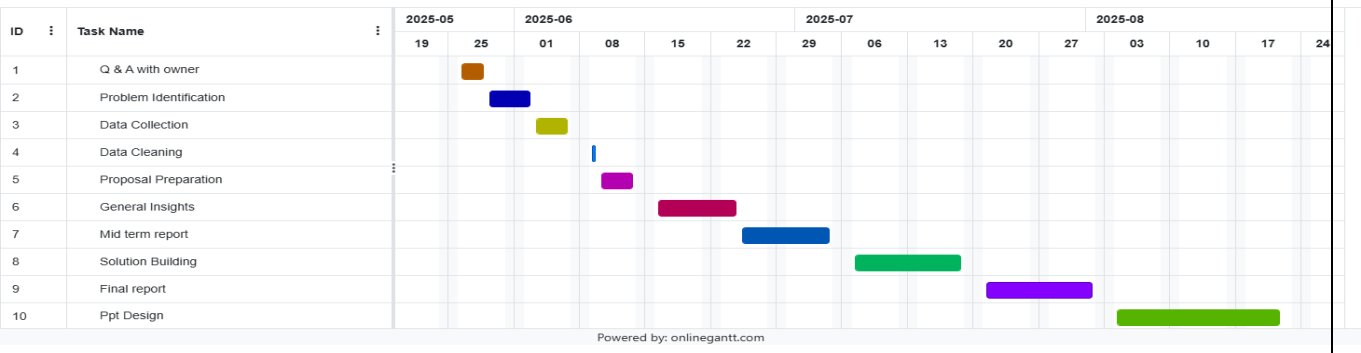


Figure 1 Expected timeline for completion of project.

Expected Outcome

Through performing thorough analysis in the areas of inventory, sales, credit, and customer behavior, the project should yield significantly effective results for the business. From analyzing the inventory, slow-moving and low-performing items will be replaced by high-margin and fast-moving items, and the turnover of stock will be improved while holding costs decrease. Implementation of ABC and dead stock analysis will assist the store in giving priority to necessary inventory and preventing blockage of funds in sluggish merchandise. This disciplined inventory management system is likely to minimize wastage of products, eliminate expirations, and maintain a continuous supply of necessary commodities. The trend analysis of sales and demand, along with seasonal trends and consumer purchasing habits, will facilitate more precise demand forecasting and procurement decisions. Therefore, the store will be able to manage the most desirable stock levels over the course of the entire year, particularly when it is most busy, thereby enhancing customer satisfaction and revenue. Also, credit and payment history analysis, including accounts receivable aging reports and repayment trend analysis, will enable the store to set the proper credit limits and terms of payment for the customers. It will minimize bad debts and enhance cash flow. Pricing and profitability analysis, based on profit margins, cost structures, and competitor benchmarking, will enable the business to create a versatile pricing system balancing customer affordability with business profitability. While customer segmentation and behavior analysis will enable the store to create focused marketing strategies, create customer loyalty, and minimize churn. Lastly, bringing all these findings together into real-time dashboards will provide ongoing performance monitoring and data-driven decision making. In total, the project should lead to increased operational efficiency, increased profit margins, increased financial integrity, and increased customer satisfaction, and lay the fundamental basis for long-term business expansion.